

OncoConnect

From Prototype to a Real-Patient Pilot

A roadmap for clinical, ethical and technical readiness

Prepared for academic supervisors / project mentors

THE CORE MESSAGE

The software is a strong, well-engineered prototype. Reaching safe use with real cancer patients is now mostly NON-software work: clinical validation, a clinician partner, ethics-committee approval, and a small supervised pilot. This document lays out that path in the order it must happen, and separates what the student can build from what requires the institution and clinicians.

Where the project stands today

OncoConnect is a secure neuro-oncology platform connecting patients, clinicians and laboratories: patients log symptoms and quality-of-life in English or Hindi; clinicians see structured, toxicity-graded data with workflow support; laboratories manage a worklist. The software is genuinely well built - encrypted patient data, tested code, a continuous-integration pipeline, and candid governance documentation.

An honest assessment separates two very different questions:

The question	Honest rating	Why
As a student / final-year engineering project	Excellent (8.5/10)	Ambitious scope, real security, thoughtful UX, honest docs
As a demonstration of a future product	Strong (7.5/10)	Demos well; documentation earns trust
As software safe for a real patient tomorrow	Low (about 3/10)	Clinical content unvalidated; no real users; no ethics/regulatory sign-off

The distance to close - stated plainly

The low real-patient score is correct and expected. It is not a criticism of the engineering; it is the normal gap between an impressive prototype and medical software. Closing it is mostly not more coding - it is clinical validation, a clinician partner, ethics approval, and evidence from a small supervised pilot.

STRATEGY

01 The key idea: narrow the use, then widen it with evidence

Full medical-device certification is a multi-year, high-cost path and is the wrong first step. There is a faster, legitimate route used by almost every digital-health tool: **narrow the intended use until it is clearly safe, obtain ethics approval for a small study, gather real evidence, then widen scope.**

For its first real use, OncoConnect should be positioned as a **symptom-diary and communication aid - explicitly not a diagnostic or dosing device.** In that framing a qualified clinician is always in the loop, the app never makes a clinical decision on its own, and real patients can use it ethically through a supervised, ethics-approved observational study.

Positioning	Risk & effort	Suitable as a first step?
Full clinical decision-support / dosing device	Very high - device certification, clinical evaluation	No - premature and expensive
Symptom diary + triage prompts + messaging (clinician in loop)	Low - non-diagnostic, supervised	Yes - the recommended first pilot

ROADMAP

02 The critical path, in order

Each step below unlocks the next. Doing them out of order wastes effort.

Step 1 - Remove or validate the unvalidated clinical content

This is the only part that could actively cause harm. The dose caps, drug-interaction list and protocol templates are illustrative content, not validated against an approved clinical database or reviewed by an oncologist. For a first pilot the correct choice is to **switch this content off** (a 'safe pilot mode'), leaving a symptom diary, triage prompts and messaging - which carry almost no clinical-safety risk. Validating the content instead is possible but slower and only needed once the doctor-facing tools are in scope.

Step 2 - Secure one clinician as a partner (the single most important step)

Everything downstream depends on a named clinician - ideally an oncologist at a hospital or medical college - who will co-own a short pilot, not merely 'review the app'. This one relationship provides clinical credibility, access to the ethics committee, and real patients. Without it the project stays a prototype. **This is not a software task and cannot be substituted by more features.**

Step 3 - Ethics committee / IRB approval for a small observational pilot

With the clinician and an institution, apply to the Institutional Ethics Committee for a small observational study, for example: 'Does a symptom-diary application improve monitoring of chemotherapy patients between clinic visits?' Roughly 15-20 patients over 3 months, with explicit written consent (already built and version-tracked in the app). Ethics review will also tighten the consent wording and data handling - a benefit, not an obstacle.

Step 4 - Independent security review before real patient data exists

Correct cryptographic primitives are necessary but not sufficient. Before any real medical record enters the system, obtain one independent security assessment. For a student this can be low-cost: a university security lab or a supervised review; a commercial penetration test is the paid option. Remediate the findings.

Step 5 - Run the pilot and collect evidence

This is what converts the prototype into evidence-backed software. Real patients, real diary-completion rates, real examples of earlier symptom escalation. The app already includes an impact dashboard and a pilot-summary export built for exactly this readout. Evidence is what every later door - publication, larger sites, eventual certification - requires.

Step 6 - Formal regulatory classification (only after pilot evidence)

If the pilot succeeds and the goal is a real product, engage a regulatory consultant for medical-device classification (in India, via CDSCO). Doing this before pilot evidence exists is premature and costly.

PLANNING

03 What each step costs, and who is needed

Step	Time	Cost	Needs people beyond the student?
1. Ship a safe (non-diagnostic) subset	Days	Nil	No - can be built now
2. Secure a clinician partner	Weeks of outreach	Nil	Yes - the decisive step
3. Ethics / IRB approval	2-3 months	Usually nil for students	Yes - clinician + institution
4. Independent security review	Weeks	Nil to high	Yes - reviewer / lab
5. Run the pilot	About 3 months	Minimal	Yes - real patients
6. Regulatory classification	6+ months	High	Yes - regulatory consultant

If you do only one thing this month

Find the clinician. Take the app to one oncologist - ideally at a medical college, where ethics-committee access is easiest - and propose a 15-20 patient symptom-diary pilot. The ask is not 'is my app good?' but 'will you help me study whether this helps your patients?'. That single agreement advances the project further than any amount of additional code.

RESPONSIBILITIES

04 Division of work: software vs. everything else

Can be delivered as software (the student's half):

- A 'safe pilot mode' that switches off unvalidated dose / interaction / protocol content, leaving diary, triage and messaging.
- A clean, de-identified study-data export and per-patient audit export for the ethics committee and analysis.
- Remediation of anything the independent security review flags.
- A plain-language patient information and consent sheet (English and Hindi) suitable for an ethics submission.
- Standardising the data model (aligned to health-data standards) so pilot data is analysis- and publication-ready.

Requires the institution and clinicians (the harder, higher-value half):

- Identifying and securing the clinician partner - nothing substitutes for this.
- Preparing and submitting the ethics-committee / IRB application through the institution.
- Real patients consenting to and using the app during the pilot.
- Engaging a regulatory consultant, when and if the project scales to a product.

THE ASK

05 What this project asks of its supervisors

This roadmap is deliberately honest about the gap so that the request is concrete and credible. Specifically, supervisor support is sought for:

- **An introduction to an oncologist or clinical department** willing to co-supervise a small symptom-diary pilot.
- **Guidance on the institution's ethics-committee process** and, where appropriate, sponsorship of the application.
- **Access to a security review** - for example through a departmental lab or a supervised student assessment.
- **Advice on framing the study** and its outcome measures so the pilot produces publishable evidence.

In return the project offers a complete, secure, tested platform, candid documentation (governance, data-protection impact assessment, implementation kit), and a clear, staged plan that manages clinical risk responsibly rather than overstating readiness.

EVIDENCE

06 Appendix: what is already built and verified

So supervisors can gauge the starting point, the following already exists in the software:

Area	Already in place
Security	AES-256-GCM encryption of all clinical data; PBKDF2 passwords; revocable sessions; role-based access; optional two-factor; audit trail
Patient app	Bilingual (English/Hindi) symptom diary, toxicity grading, quality-of-life survey, explicit 'when to call now' triage with one-tap calling, accessibility modes
Clinician app	Patient records, symptom trends, workflow support, impact analytics, governance dashboard
Consent & privacy	Versioned consent with history and withdrawal; de-identified research export; data-protection mapping (DPDP / GDPR)
Engineering	20 automated tests passing; continuous-integration pipeline; observability; documented deployment
Documentation	Governance policy, data-protection impact assessment, clinic implementation kit, full project dossier

This document describes a responsible, staged plan to move OncoConnect from a well-built prototype toward evidence-based use with real patients. It does not claim regulatory or clinical clearance, and it treats patient safety and ethics as the gating requirements they are.